Example lecture

1. Compute the predictor $\hat{x}_4$ for the time series $x_1 = 97$, $x_2 = 102$, $x_3 = 101$ from the stochastic process with $R(k) = 1/(1+k^2)$. Also compute 95% confidence bounds for x_4.

We have $\hat{\mu} = (97 + 102 + 101)/3 = 100$. $Y = (-3, 2, 1)$, $\Gamma \cong \begin{pmatrix} 1 & 0.5 & 0.2 \\ 0.5 & 1 & 0.5 \\ 0.2 & 0.5 & 1 \end{pmatrix}$, $\Gamma^{-1} = \begin{pmatrix} 1.34 & -0.71 & 0.09 \\ -0.71 & 1.71 & -0.71 \\ 0.09 & -0.71 & 1.34 \end{pmatrix}$, $b_1 = \begin{pmatrix} 0.1 \\ 0.2 \\ 0.5 \end{pmatrix}$, $a = \Gamma^{-1} b_1 = \begin{pmatrix} 0.036 \\ -0.086 \\ 0.53 \end{pmatrix}$, $\hat{y}_4 = a^T Y = 0.26$, $\hat{x}_4 = \hat{\mu} + \hat{y}_4 = 100 + 0.26 = 100.26$, $\hat{\sigma}_4 = 0.86$, The 95% confidence interval is $(98.56, 101.94)$.

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x=c(97,102,101); xpred=rep(0,3); xsig=rep(0,3); givenCov=function(k) { return(1/(1+k*k)); } ; n=length(x); mu=mean(x); print(mu); y=x-mu; print(y); Gamma=matrix(0,n,n); for(i in (1:n)) for(j in (1:n)) Gamma[i,j]=givenCov(i-j); print(Gamma); print(solve(Gamma)); for(h in (1:1)) { bh=rep(0,n); for(i in (1:n)) bh[i]=givenCov(n+h-i); print(bh); a=solve(Gamma) %*% bh; print(a); ynh=y%*%a; print(ynh); xpred[h]=mu+ynh; xsig[h]=sqrt(givenCov(0)-bh %*% solve(Gamma) %*% bh); print(xpred[h]); print(xsig[h]); print(xpred[h]+xsig[h]*qnorm(1-0.05/2)*c(-1,1)); }
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2. Compute the predictor $\hat{x}_8$ for the time series $x_5 = 14$, $x_6 = 11$, $x_7 = 13$ from the stochastic process with mean 12 and $R(k) = 1/(1 + 2\sqrt{|k|})$. Also compute 95% confidence interval for x_8.

We have $\mu = 12$. $Y = (2, -1, 1)$, $\Gamma \cong \begin{pmatrix} 1 & 0.33 & 0.26 \\ 0.33 & 1 & 0.33 \\ 0.26 & 0.33 & 1 \end{pmatrix}$ $\Gamma^{-1} = \begin{pmatrix} 1.16 & -0.32 & -0.196 \\ -0.32 & 1.21 & -0.32 \\ -0.196 & -0.32 & 1.16 \end{pmatrix}$, $b_1 = \begin{pmatrix} 0.22 \\ 0.26 \\ 0.33 \end{pmatrix}$, $a = \Gamma^{-1} b_1 = \begin{pmatrix} 0.11 \\ 0.14 \\ 0.26 \end{pmatrix}$, $\hat{y}_8 = a^T Y = 0.34$, $\hat{x}_8 = \hat{\mu} + \hat{y}_4 = 12 + 0.34 = 12.34$, $\hat{\sigma}_8 = 0.92$, The 95% confidence interval is $(10.53, 14.15)$.

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x=c(14,11,13); xpred=rep(0,3); xsig=rep(0,3); givenCov=function(k) { return(1/(1+2*sqrt(abs(k)))); } ; n=length(x); mu=12; print(mu); y=x-mu; print(y); Gamma=matrix(0,n,n); for(i in (1:n)) for(j in (1:n)) Gamma[i,j]=givenCov(i-j); print(Gamma); print(solve(Gamma)); for(h in (1:1)) { bh=rep(0,n); for(i in (1:n)) bh[i]=givenCov(n+h-i); print(bh); a=solve(Gamma) %*% bh; print(a); ynh=y%*%a; print(ynh); xpred[h]=mu+ynh; xsig[h]=sqrt(givenCov(0)-bh %*% solve(Gamma) %*% bh); print(xpred[h]); print(xsig[h]); print(xpred[h]+xsig[h]*qnorm(1-0.05/2)*c(-1,1)); }
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3. Compute the estimators $\hat{R}(0)$, $\hat{R}(1)$, $\hat{R}(2)$ for the time series $x_1 = 12$, $x_2 = 7$, $x_3 = 11$.

Direct calculus gives $\hat{R}(0) = 4.667$, $\hat{R}(1) = -3.0$ and $\hat{R}(2) = 0.667$.

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x=c(12, 7, 11); n=length(x); mu=sum(x)/n; y=x-mu; covg=rep(0,n); corrg=rep(0,n); for(i in (1:n)) { covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n; corrg[i]=covg[i]/covg[1]; } ; print(covg[c(0,1,2)+1])
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4. Consider the time series 11, 13, 11, 15, 10. Compute estimators of the correlation function $\hat{\rho}(1)$ and $\hat{\rho}(2)$.

Direct calculus gives $\hat{\rho}(1) = -0.6875$ and $\hat{\rho}(2) = 0.375$.

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x=c(11, 13, 11, 15, 10); n=length(x); mu=sum(x)/n; y=x-mu; covg=rep(0,n); corrg=rep(0,n); for(i in (1:n)) { covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n; corrg[i]=covg[i]/covg[1]; } ; print(corrg[c(1,2)+1])
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